

# Austin Jones

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**Objective:** Software Engineer with experience in Embedded Software and Cloud Development looking for cool problems to solve.

## Skills

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Cloud Technologies: Docker, K8s, AWS, GitHub Actions, gRPC, Postgres  
Embedded Technologies: ARM MCUs, Xilinx FPGAs/MPSoCs  
Project Management: GitHub, Jira, Azure, Agile  
Languages: C, Rust, C#, C++, .NET, Shell, Python, VHDL

## Work Experience

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Current –  
Jan 2024

**Uster** – Knoxville, TN

Work to make applications and solutions to facilitate Textile Grading devices.

### **Software Engineer:**

- Own an application for aggregating and statusing state across large sites.
  - Own the project from customer spec to delivery of 1.0.
  - Design a UI/UX to show 500 parameters from 50 machines.
  - Semi-real time updates from 50 machines with smooth UI.
  - Develop lab infra to allow for a swarm of virtual devices to exercise the application away from customer site.
  - Accidentally write a UI framework.
    - \* Converts a JSON UI spec into WPF screens.
    - \* Allows for easy maintenance of 50 ui screens and hundreds of files.
    - \* Consolidates field validation, styling, and data binding per field type..
- Inherit and support utility software for manufacture of products.
  - Solely inherit a 5 year old application from another team of 5.
  - Support its use on site for ongoing manufacture of company products.
  - Assess work needed to go from MVP hybrid use to full use.
  - Own this work and its pre-release validation.
  - Over see the validation and deployment of the new procedure.
- Provide software support for implementing research algorithms into production software.
- Developed algorithms to align data product across instruments.
- Organize and run efforts relating to company purpose and employee development.
  - Spearhead a grass roots program for employee recognition.
  - Start a group to help employees gain autonomy for projects they want to happen.
- Take on a summer intern to get an intern program started.
  - Plan and manage work of the intern.
  - Learn how to let people fail to learn.
  - Organize lunch and learns, knowledge transfer sessions, and general intern things.

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| Nov 2023 -<br>Jan 2022 | <p><b>PolySign - Remote</b></p> <p>Collaborated with many teams across multiple products and technologies to build out features and products for a Digital Asset Custodian in the quickly evolving Cryptocurrency market.</p> <p><b>Senior Software Engineer:</b> Jan 2023 - Nov 2023</p> <ul style="list-style-type: none"> <li>Contributed to Escrow feature of Python microservice backend to meet new customer requirements: <ul style="list-style-type: none"> <li>Redesigned feature data model and control flow to increase robustness.</li> <li>Designed a new data flow to allow for more flexibility in the feature.</li> </ul> </li> <li>Implemented an end-to-end POC for an unreleased feature across multiple projects.</li> <li>Worked on legacy C/Rust codebase to support features in other products.</li> <li>Performed monthly Dev-on-Call rotations debugging live issues across multiple environments including: dev, testing, pre-prod, and prod.</li> <li>Planned, hosted, and participated in a company wide Hackathon.</li> <li>Organized and hosted a weekly engineering reading club.</li> </ul> <p><b>Software Engineer:</b> Jan 2022 - Dec 2022</p> <ul style="list-style-type: none"> <li>Helped stand up a green field project in Rust aimed at performing fast, distributed multiparty compute signing: <ul style="list-style-type: none"> <li>Designed fully asynchronous architecture for distributed message signing increasing efficiency and robustness.</li> <li>Overhauled network stack to use Tokio's Tonic gRPC over VirtioVsock Sockets increasing dev velocity and code cleanliness.</li> <li>Redesigned CI/CD Pipeline for multi-repo project to minimize redundancy in builds, regression testing, and integration testing.</li> </ul> </li> <li>Integrated with main product as a POC.</li> </ul> |
| Aug 2021 -<br>May 2021 | <p><b>Garmin International - Olathe, KS</b></p> <p><i>Embedded Software Intern</i></p> <p>Developed code for microcontroller to control power on logic for an unreleased product.</p> <ul style="list-style-type: none"> <li>Utilized MCU to monitor main board status and determine the power state of the board.</li> <li>Referenced PCB schematics and MCU documentation to integrate software and hardware.</li> <li>Gained strong familiarity with ARM/STM microcontrollers.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Nov 2021 -<br>Jan 2019 | <p><b>Siemens Molecular Imaging - Knoxville, TN</b></p> <p><i>Electrical R&amp;D Intern</i></p> <p>Conduct projects, both individually and collaboratively, to provide value to the ER&amp;D team:</p> <ul style="list-style-type: none"> <li>Presented findings from work to a large technical audience to demonstrate the validity of a new system architecture.</li> <li>Long term projects: <ul style="list-style-type: none"> <li>Utilize a Xilinx MPSoC to test limitations of ARM Core processing vs an FPGA implementation to assess cost reductions.</li> <li>Used multiple Raspberry Pis with off the shelf networking hardware to conduct data path tests and compare the bandwidth with current custom solutions.</li> <li>Work with a team of engineers using GitLab to write various driver code for a custom Linux image.</li> <li>Developed Firmware, Embedded Software, and Application Software for a unit test fixture that is used during production of PET electronics.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Personal Projects

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| Jan 2025 | <b>ABC Notepad</b><br>An app that is meant to help with a CBT journaling style. Made in React Native with the Expo Router. Built and deployed with EAS.                                                                                                  |
| Dec 2024 | <b>My blog</b><br>A place for me to put posts and articles. I mainly write about personally philosophy. Made with Svelte 5.                                                                                                                              |
| May 2024 | <b>Austin's Arcade</b><br>A collection of JavaScript games. They are mostly remakes of classic arcade games. They've been an exercise in me working fast and shipping.                                                                                   |
| Dec 2023 | <b>Connect Four Bot</b><br>A connect four bot written in Rust. It implements basic alpha-beta minimax. At a shallow depth it's formidable - deeper depths it is impossible.                                                                              |
| Jan 2021 | <b>Light Sign with Web Server Control</b><br>Controlled an LED Array to display text specified on a web page. Custom web server written in Rust hosted control page and sent data over UART. Arduino received UART data and displayed data on LED Array. |

## Education

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| <b>University of Tennessee Knoxville</b>    |                                                                                                                                                                                                                                                                               | Anticipated Graduation: Dec 2021 |
| Bachelor of Science in Computer Engineering |                                                                                                                                                                                                                                                                               | Gpa: 4.0/4.0                     |
| Dec 2021                                    | <b>Engineering Professional Practice Office</b> - University of Tennessee                                                                                                                                                                                                     |                                  |
| Aug 2019                                    | <i>Student Ambassador</i><br>Assist UT engineering students to improve resumes, interview/interview person skills, and overall professional skill. Work with other ambassadors to facilitate networking vents and career fairs.                                               |                                  |
| Jan 2021                                    | <b>Light Sign with Web Server Control</b> - Personal Project<br>Controlled an LED Array to display text specified on a web page.                                                                                                                                              |                                  |
| Apr 2019                                    | <b>Password Manager Browser Extension</b> - University of Tennessee<br><i>Data Structures and Algorithms Final Project</i><br>Collaborated with a group of peers using Github to create a functional Chrome extension that manages a user's passwords for multiple sites.     |                                  |
| Apr 2019                                    | <b>Render Component for Firmware Based Pong</b> - University of Tennessee<br><i>Digital Systems Design Final Project</i><br>Reimplemented the retro game Pong on an FPGA using VHDL. Component was implemented to read the game state and render all game objects and scores. |                                  |

## Scholarships and Honors

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| May 2019           | Gonzalez Family Awards for Outstanding Computer Engineering Junior |
| Aug 2019 - Current | S. T. Harris Scholarship                                           |
| Aug 2019 - Current | Herschel C. and Louise Runnion Brand Engineering Scholarship       |